UNIT 8 • QUADRATICS

Activity 8.2: Features of Parabolas

A

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question about the quadratic function. Show your work.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1 Find the axis of symmetry of $y = x^2 - 6x + 5$.

Answer: _____ Match? ☐ Yes ☐ No

2 Find the vertex of $y = x^2 - 4x + 1$.

Answer: _____ Match? ☐ Yes ☐ No

3 Find the y-intercept of $y = 2x^2 - 3x + 5$.

Answer: _____ Match? ☐ Yes ☐ No

4 Does $y = -3x^2 + 1$ open up or down? Does it have a maximum or a minimum?

Answer: _____ Match? ☐ Yes ☐ No

5 Find the x-intercepts of $y = x^2 - 4x - 5$.

Answer: _____ Match? ☐ Yes ☐ No

6 Find the vertex of $y = -x^2 + 2x + 3$.

Answer: _____ Match? ☐ Yes ☐ No

7 A ball's height in feet after t seconds is $h = -16t^2 + 64t$. At what time is the ball at its maximum height?

Answer: _____ Match? ☐ Yes ☐ No

8 How many x-intercepts does $y = x^2 + 4x + 5$ have?

Answer: _____ Match? ☐ Yes ☐ No

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Activity 8.2: Features of Parabolas

B

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question about the quadratic function. Show your work.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1 Find the axis of symmetry of $y = 2x^2 - 12x + 7$.

Answer: _____ Match? ☐ Yes ☐ No

2 Find the vertex of $y = 2x^2 - 8x + 5$.

Answer: _____ Match? ☐ Yes ☐ No

3 Find the y-intercept of $y = -x^2 + 4x + 5$.

Answer: _____ Match? ☐ Yes ☐ No

4 Does $y = 5 - 2x^2$ open up or down? Does it have a maximum or a minimum?

Answer: _____ Match? ☐ Yes ☐ No

5 Find the x-intercepts of $y = 2x^2 - 8x - 10$.

Answer: _____ Match? ☐ Yes ☐ No

6 Find the vertex of $y = -2(x - 1)^2 + 4$.

Answer: _____ Match? ☐ Yes ☐ No

7 A ball's height in feet after t seconds is $h = -16t^2 + 64t + 5$. At what time is the ball at its maximum height?

Answer: _____ Match? ☐ Yes ☐ No

8 How many x-intercepts does $y = 2x^2 + x + 3$ have?

Answer: _____ Match? ☐ Yes ☐ No